

AI-DRIVEN EVENT MONITORING SYSTEM USING COMPUTER VISION AND VISION-LANGUAGE MODELS

Abstract:

Large-scale events require continuous situational awareness, rapid incident triage, and coordinated response across multiple stakeholder roles. This project presents an integrated AI-based event monitoring and response platform that unifies live surveillance analytics, incident reporting, volunteer coordination, and attendee assistance within a single full-stack system. The backend combines Express, MongoDB, and Socket.IO for realtime issue tracking, alert propagation, and role-based workflow management, while the frontend uses React and Vite to provide dedicated dashboards for admins, volunteers, and attendees. On the AI layer, the system uses YOLOv8 models for real-time fire and face/object detection, CLIP-based semantic classification for violence detection, and deep face-recognition embeddings for identity search. A chatbot further summarizes preloaded event logs to generate operational insights. The architecture is designed to convert AI detections into actionable alerts, assign them to authorized responders, and track resolution status end-to-end. By bridging computer vision, vision-language modeling, identity verification, and human coordination, the platform provides a practical framework for safer, faster, and more organized event management.

Keywords:

Event monitoring, YOLOv8, computer vision, real-time surveillance, incident management, face recognition, CLIP, geolocation tracking

Team No: 18

Domain of the Project: Artificial Intelligence & Machine Learning

Team Members:

S. No.	Roll No	Student Name	Phone no
1.	23BD1A1216	GAMPA ANUPAMA	94928 05265
2.	23BD1A1231	KODIPYAKA SUJANA GUPTA	96666 69284
3.	23BD1A1239	MUDIGA SHIVANI	70134 18370

Faculty Supervisor

Project Coordinator